

REFLECTION FOR HW13A

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Design Rationale

From the beginning, I wanted the calculator application to have a clean structure instead of putting all logic into a single class. Therefore, I separated the project into two main parts: the UI layer and the calculation logic layer. The CalculatorFrame class is responsible for building the graphical user interface and handling button events, while CalculatorLogic handles arithmetic operations and result formatting.

For the user interface, I used Java Swing together with GridBagLayout to create a calculator layout similar to a real calculator application. One design detail I focused on was making the "=" button span two columns to improve the visual appearance and make the layout more user-friendly.

I also wanted the application to be easy to maintain and extend later. Because of that, each method in CalculatorFrame has only one specific responsibility, such as handling number input, operator input, clearing the display, deleting characters, or calculating results.

Main Functionalities

1. Basic Arithmetic Operations:

The calculator supports addition, subtraction, multiplication, division, and modulo operations.

2. Display and Input Handling:

Users can enter numbers and decimal values using on-screen buttons. The display area shows the current input and the final result.

3. Control Buttons:

The DEL button removes the last entered character, while the C button clears the entire calculator state.

4. Error Handling:

The application properly handles invalid cases such as division by zero, incomplete expressions, invalid numbers, and other illegal operations using dialog error messages.

Implementation

The project is divided into two packages: ui and logic.

In the logic layer, CalculatorLogic contains methods for performing arithmetic calculations and formatting results. Separating this logic from the interface makes the program easier to manage and avoids mixing GUI code with calculation code.

In the UI layer, CalculatorFrame extends JFrame and creates all Swing components such as JTextField, JButton, JPanel, and layouts. Event handling is implemented using ActionListener for all buttons. GridBagLayout is used to organize buttons neatly and create a larger "=" button similar to a standard calculator interface.

Overall, this exercise helped me practice Java Swing GUI development, event-driven programming, layout management, and object-oriented design principles.